AWS Cloud Architecture & Free Tier Implementation Guide

Optimal Zero-Cost AWS Cloud Services for InduSense Industrial Maintenance Intelligence

CLOUD PROVIDER: Amazon Web Services (AWS) **COST TARGET:** $0.00 / Month (100% Free Tier) **TARGET PLATFORM:** InduSense Predictive RAG **DATE:** November 2026

1. Executive Summary: Why Cloud Computing for InduSense?

InduSense is an industrial multimodal platform bridging high-frequency sensor telemetry (AI4I, FEMTO bearings, Hydraulic systems) with technical documentation intelligence (OEM repair manuals, electrical schematics). To transition InduSense from a local development prototype into an enterprise-ready, production-grade cloud solution without incurring billing costs, Amazon Web Services (AWS) offers a rich combination of **Always Free** and **12-Month Free Tier** services.

> **Guaranteed Zero-Cost Architecture:** Every AWS service selected in this guide operates strictly within AWS Free Tier allowances. By combining serverless computing, object storage, and managed edge delivery, the entire InduSense platform can process thousands of telemetry diagnosis runs, store hundreds of OEM manuals, and host an active web portal for **$0.00/month**.

2. Complete AWS Free Tier Service Mapping Matrix

AWS Service	InduSense Pipeline Stage	Official Free Tier Quota	Free Tier Type	Monthly Cost
AWS Lambda	1. Predict & 3. CEA Alignment	1,000,000 requests/mo + 3.2M sec compute	ALWAYS FREE	**$0.00**
Amazon S3	2. Retrieve (OEM Manuals Storage)	5 GB Standard Storage + 20,000 GETs	12-MO FREE	**$0.00**
Amazon DynamoDB	6. Explain & Audit Trail Log	25 GB Storage + 25 WCU / 25 RCU	ALWAYS FREE	**$0.00**
Amazon CloudFront + S3	Vite React Frontend Web Hosting	1 TB Data Transfer/mo + 10M HTTP requests	ALWAYS FREE	**$0.00**
Amazon Cognito	RBAC Auth (Engineer/Tech/Admin)	50,000 Monthly Active Users (MAUs)	ALWAYS FREE	**$0.00**
Amazon API Gateway	REST API Orchestration Gateway	1,000,000 REST API calls/month	12-MO FREE	**$0.00**
Amazon EC2 (Optional)	FastAPI & Vector Database Hosting	750 hours/mo (t2.micro / t3.micro)	12-MO FREE	**$0.00**
Amazon CloudWatch	Telemetry Monitoring & SLAs	10 Custom Metrics + 5 GB Logs	ALWAYS FREE	**$0.00**

AWS Service	InduSense Pipeline Stage	Official Free Tier Quota	Free Tier Type	Monthly Cost
Amazon SNS	Critical Fault Emergency Alerts	1,000,000 Push + 1,000 Emails/month	ALWAYS FREE	**$0.00**

3. In-Depth Technical Breakdown of Each Recommended Service

1. AWS Lambda — Serverless Telemetry Processing & Inference

ALWAYS FREE: 1M INVOCATIONS / MO

Why it is useful for InduSense: Industrial sensor telemetry (such as CNC spindle RPM, bearing vibration kurtosis, or hydraulic valve switching latency) arrives intermittently or in batches. With AWS Lambda, you do not pay for idle servers. Lambda executes the trained scikit-learn / XGBoost model (packaged with ONNX Runtime or a lightweight container) within milliseconds.

- **Role in InduSense:** Runs the *Predict* stage (classifying failure modes such as HDF, TWF, or BPFO bearing defect) and the *Condition-Evidence Alignment (CEA)* calculation.
- **Free Tier Allocation:** 1,000,000 free requests per month, plus 3.2 million seconds of compute time per month. That allows running over 30,000 full diagnostic cases every single day at zero cost.

2. Amazon S3 — Technical Manuals & Telemetry Data Lake

12 MONTHS FREE: 5 GB STORAGE

Why it is useful for InduSense: InduSense requires storing both structured telemetry CSV files and unstructured OEM technical documents (PDF maintenance manuals, wiring schematics, and disassembly procedures). S3 provides 99.999999999% (11 9s) durability.

- **Role in InduSense:** Stores technical manuals like `06_USER_MANUAL_INDUSENSE_PLATFORM.pdf`, `07_MANUAL_CNC_SPINDLE_MAINTENANCE_SOP.pdf`, and raw telemetry CSV files. Serves as the document repository for RAG embedding extraction.
- **Free Tier Allocation:** 5 GB of standard object storage, 20,000 GET requests, and 2,000 PUT requests per month. 5 GB can easily store over 1,500 full-length technical equipment manuals.

3. Amazon DynamoDB — Real-time Case Workspace & Tamper-Evident Audit Trail

ALWAYS FREE: 25 GB NOSQL STORAGE

Why it is useful for InduSense: Industrial maintenance mandates an immutable, timestamped audit trail of every review, sign-off, and engineering override. DynamoDB provides single-digit millisecond latency at any scale without managing database clusters.

- **Role in InduSense:** Stores diagnosis case objects, sensor telemetry snapshots, versioned technician sign-offs, and audit event logs (`CASE_INITIALIZED`, `ALIGNMENT_EVALUATED`, `OVERRIDE_RECORDED`).
- **Free Tier Allocation:** 25 GB of storage, 25 provisioned Write Capacity Units (WCU), and 25 provisioned Read Capacity Units (RCU) — *Always Free*. Capable of handling over 200 million read/write requests per month.

4. Amazon CloudFront + S3 — Ultra-Fast React Frontend Web Hosting

ALWAYS FREE: 1 TB BANDWIDTH / MO

Why it is useful for InduSense: The InduSense web user interface (built with Vite React and Tailwind CSS) must load instantly on plant tablets and desktop workstations. CloudFront distributes the frontend across global edge locations with SSL/TLS encryption.

- **Role in InduSense:** Serves the compiled frontend build (`dist/`), routing pages seamlessly across Operations, Case Workspace, Knowledge Base, Cloud Router, and Settings.
- **Free Tier Allocation:** 1,000 GB (1 TB) of data transfer out per month, plus 10,000,000 HTTP/HTTPS requests per month forever. You will never pay for frontend web traffic.

5. Amazon Cognito — Enterprise Role-Based Access Control (RBAC)

ALWAYS FREE: 50,000 MONTHLY ACTIVE USERS

Why it is useful for InduSense: InduSense enforces strict role separation between **Lead Reliability Engineers** (audit and override authority), **Maintenance Technicians** (inspection checklists and defect submission), and **Administrators**. Cognito handles secure JWT token issuance, sign-in, and password hashing.

- **Role in InduSense:** Authenticates operators, attaches role claims (`engineer`, `technician`, `admin`) to API requests, and secures sensitive override actions.
- **Free Tier Allocation:** 50,000 Monthly Active Users (MAUs) completely free, indefinitely.

6. Amazon API Gateway — Unified Microservice Entrypoint

12 MONTHS FREE: 1,000,000 API CALLS / MO

Why it is useful for InduSense: Acts as the single reverse proxy and API entrypoint for the frontend application. It securely validates Cognito JWT tokens, limits request throttling, and dispatches traffic to Lambda functions or EC2 endpoints.

- **Role in InduSense:** Routes `/api/cases`, `/api/telemetry`, `/api/documents`, and `/api/cases/{id}/review`. Handles CORS headers automatically.
- **Free Tier Allocation:** 1,000,000 REST API calls per month for 12 months.

7. Amazon EC2 (t2.micro / t3.micro) — All-in-One Backend Alternative

12 MONTHS FREE: 750 HOURS / MONTH (CONTINUOUS 24/7)

Why it is useful for InduSense: If you prefer to deploy the existing Python FastAPI application and Chroma/SQLite database without converting them to serverless Lambda functions, a single EC2 `t2.micro` (or `t3.micro`) provides 1 vCPU and 1 GB of RAM running 24 hours a day, 7 days a week.

- **Role in InduSense:** Runs `uvicorn server.main:app`, local vector similarity search, and model inference checkpoints directly inside a virtual Linux environment.
- **Free Tier Allocation:** 750 hours per month (enough to run 1 instance uninterrupted 31 days a month) for 12 months, plus 30 GB of EBS general purpose SSD storage.

8. Amazon CloudWatch & Amazon SNS — Real-Time Monitoring & Critical Alerts

ALWAYS FREE: 10 METRICS + 1M PUSH ALERTS

Why it is useful for InduSense: Critical machine failures (like Heat Dissipation Failure on CNC-M102 or outer race spalling on Bearing B301) require immediate notification to on-call maintenance teams.

- **Role in InduSense:** CloudWatch tracks pipeline latency (predict, retrieve, align duration SLAs). Amazon SNS dispatches automated email/SMS notifications whenever an *Escalated: Evidence Conflict* state is flagged.
- **Free Tier Allocation:** CloudWatch: 10 custom metrics, 10 alarms, 5 GB log ingestion (Always Free). SNS: 1,000 email alerts and 1,000,000 mobile push notifications per month (Always Free).

4. Step-by-Step Deployment Strategy to Stay 100% Free

1. **Set an AWS Zero-Spend Budget Alert:** Go to *AWS Budgets* → create a budget with an alert threshold of **$0.01**. If any service ever exceeds free tier usage, AWS will immediately email you before charges accumulate.
2. **Deploy Frontend via CloudFront + S3:** Run `npm run build` in the frontend folder, upload `dist/` to an S3 bucket configured for static website hosting, and link it to an Amazon CloudFront distribution.
3. **Deploy Backend on EC2 t2.micro or Serverless Lambda:**
   - *Option A (Easiest):* Launch a free Amazon Linux 2023 `t2.micro` instance, clone the project, and run with systemd / PM2. Zero code changes required!
   - *Option B (Pure Serverless):* Wrap FastAPI in `Mangum` and deploy as an AWS Lambda function behind HTTP API Gateway.
4. **Store Manuals in S3 Bucket:** Upload the generated PDF manuals from `demo_data_and_user_manuals/` to an S3 bucket named `indusense-technical-manuals`.

Conclusion: With this architecture, InduSense operates as a scalable, fault-tolerant, enterprise industrial intelligence platform on AWS with **$0.00 infrastructure cost**.